Practice Quiz 09C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Inverse images of intersections; diagnosing an induction proof.

1. (a) Let $f : A \to B$ be a function and let $Y, Z \subseteq B$. Prove

$$f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z).$$

Solution Let $x \in A$ be arbitrary. From the definition of inverse image,

$$\begin{aligned} x \in f^{-1}(Y \cap Z) &\iff f(x) \in Y \cap Z \\ &\iff f(x) \in Y \text{ and } f(x) \in Z \\ &\iff x \in f^{-1}(Y) \text{ and } x \in f^{-1}(Z) \\ &\iff x \in f^{-1}(Y) \cap f^{-1}(Z). \end{aligned}$$

Both sides are subsets of A, and they have the same elements, so they are equal. ∎

(b) A student claims that $\sum_{i=0}^{n} 2^i = 2^{n+1}$ for every integer $n \geq 0$ and writes: *Assume the formula holds at n. Adding 2^{n+1} gives 2^{n+2}, so it holds at $n+1$. Thus induction proves the claim.* Identify the gap by checking a base case, and state a corrected formula.

Solution The proof never verifies a base case. At $n = 0$, the left side is $2^0 = 1$, while the claimed right side is $2^1 = 2$. Thus the claimed formula is false. A valid implication from the nth case to the next case does not replace a base case.

The corrected formula is

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1 \qquad (n \geq 0).$$

Its base case is $1 = 2 - 1$, and it is the geometric sum formula with ratio 2. ∎

Solutions revised using GPT-6 Astra in the Codex harness.